Project 2 — Milestone

Phase A · before data collection · Module 2

Project 2 milestone

Phase A in: Project 2 — Estimating the Truth

What this checkpoint is for

This check lets you plan and defend your study **before** you visit sampled items and compute a confidence interval (Phase B on the Project 2 page). That way we catch a bad frame, vague variable, or non-SRS plan early. Do not collect data, graph your sample, or compute a 95% CI yet — those are Phase B.

Why fix population, variable, and SRS IDs before you collect?

In Module 0, Read 0, the five-step statistical process puts planning before collecting. Step 2 is to decide what you will sample, how you will sample it, and how you will analyze it — *before* Step 3 (gather data from the world we observe). Phase A of this milestone is your Step 2 for Project 2: you fix the finite population (N), the variable, sample size n, and the list of SRS IDs so your later interval is tied to a design stated in advance.

Module 2, Read 0 previews the same idea for estimation: name the population, unit, frame, SRS plan, n, and whether you will estimate a mean or proportion (and use the FPC when N is finite) before you have sample values. If you wait until after data collection to pick the population or sampling method, you blur "what I planned to measure" with "what was convenient to measure," and the story for your CI weakens.

Phase A — complete in the notebook

Step	Task
1	Choose a listable population you can randomly sample from and visit (or inspect) each sampled item.
2	Choose the parameter and variable — binary vs numerical; proportion vs mean.
3	Choose sample size n (10 objects, or the largest feasible in 1–2 hours).
4	Draw an SRS — use a random number generator to list n distinct IDs from 1 to N.

Step 1. Choose a finite population you can list

Think of a population you could list, randomly sample from the list, and visit (or inspect) each sampled item to collect one variable — either a number or a binary outcome.

It is hard to randomly sample people without university research approval and without a complete contact list for the population. For these reasons, I recommend a population of non-human objects. For example:

- Number all parking spaces in a lot (e.g. from a Google Maps image), randomly select 10 spaces, and record one variable at each space (occupied or not? truck or not? white or not? model year?).
- Sample from a numbered list of items on a website, as in the sample project.

Your turn.

My population is ____, which has $N =$ ________ objects.

Step 2. Choose the parameter and variable

Once you have a population, decide what **mean** or **proportion** you want to estimate. Identify the **variable** you will collect for each sampled object.

Your choices in step 2 commit you to a mean CI (`t.test` + FPC when needed) or a proportion CI (`binom.test` / `prop.test` + FPC when needed) in Phase B.

Your turn.

My variable is ________________________________, which is binary categorical numerical (circle one),
so I'll be estimating a proportion mean (circle one).

Step 3. Choose a sample size

Larger samples give more precision, but collecting data takes time. In professional work, a sample size calculation balances time and precision. For this project, collect data from 10 objects or the largest number you can feasibly visit in 1–2 hours.

Your turn.

I think the largest number of objects I could feasibly visit and collect information for is $n =$ ________.

Step 4. Draw a simple random sample

Use a random number generator to select n distinct items from 1 to N.

Rossman Chance random number generator or

`sample(1:50, 6)` in R returns a random set of 6 numbers from 1 to 50.

Your turn.

The n items I need to visit are: ____

____

What to turn in to Canvas (optional)

Submit your answers for steps **1–4**:

1. Population description and N
2. Variable type and whether you will estimate a mean or proportion
3. Planned sample size n
4. List of n distinct SRS IDs from 1 to N

After the milestone

Complete **Phase B** on the Project 2 page:

1. Collect your data (step 5).
2. Compute your point estimate (step 6).
3. Check conditions and compute a 95% CI in R (step 7).
4. Interpret the interval (step 8).
5. One-page summary + notebook using the Project 2 rubric.